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Enhancing Disturbance Rejection in Minisatellite Attitude Control: A Comparative Study

Emre Sayin

Aeronautical Engineering Department
Istanbul Technical University
Istanbul, Türkiye
emre.sayin@itu.edu.tr

Ismail Bayezit

Aeronautical Engineering Department
Istanbul Technical University
Istanbul, Türkiye
bayezit@itu.edu.tr

Abstract—This research explores the application and performance of the Linear Quadratic Regulator (LQR) and Model Predictive Controller (MPC) in satellite attitude control. The study involves deriving the satellite equations of motion by defining the kinematics and dynamics equations. The nonlinear state-space equations are constructed, and dimension reduction linearization is used to make the system controllable. The theory of the controllers is explained, and simulations are conducted to evaluate their effectiveness. The simulations assess the impact of control input saturation on LQR controller and show the time response of the designed MPC controller. The comparison of the two controllers focuses on their disturbance rejection performance. Results demonstrate that the disturbance performance of the MPC controller is significantly better than LQR controller and that the constrained MPC controller can dampen step disturbance, while the LQR controller has no impact on rejection. Overall, this study contributes to the growing knowledge base on satellite attitude control and provides valuable insights for improving its efficiency and effectiveness.

Index Terms—model predictive control, linear quadratic regulator, state-space representation, spacecraft attitude control

I. INTRODUCTION

It is crucial for a spacecraft to be oriented in a specific direction in space. This might be required for the missions such as military observations [1], climate change observation, space observation [2], [3], and so on. The satellite needs to be oriented to the given direction autonomously. Thus, most feedback control systems are utilized to autonomously attitude control design.

There are various methods to control the attitude of a spacecraft. One of the advanced methods used in attitude control is Linear Quadratic Regulator (LQR) control. This method is based on the solution of an optimization problem that can be used to minimize reference tracking error, the fuel of the spacecraft, and so on. For the satellite attitude control, both experimental and simulation studies are presented in the literature. In [4], different control methods are designed for ESTCube2/3 missions in which, one of the methods is LQR control. Navabi and Hosseini design a quaternion-based attitude controller for a reaction wheel-actuated satellite. They use an input-output feedback linearized system and design an LQR controller for a linearized system [5]. Another study focuses on designing of sliding mode and LQR controllers for disturbance rejection on attitude control. They tune the parameter of the controllers by using the Particle Swarm Optimization method

and compare them by using specific performance criteria [6]. Yang design an LQR controller by using a reduced quaternion model, in which the system becomes fully controllable [7].

A different method of satellite attitude control is called Model Predictive Controller (MPC). It is based on solving optimization problems just like the LQR. However, MPC has the capability of constraint handling which is important for autonomous control cases. Studies also were proposed on attitude control of the satellites by MPC. In [8], an MPC controller is designed for a CubeSat nanosatellite. They construct a linear time-variant model to control the attitude by using MPC. They also elaborate the usage of the Extended Kalman filter. In [9], attitude control in the presence of magnetorquers is investigated. They use MPC on attitude control and compare the results with a well-known controller type, PD controller. Mirshams and Khosrojerdi developed a control method called "Tube-Based MPC" for an underactuated satellite that lost the ability to move around one axis [10]. More studies on MPC controller based attitude control can be found in [11], [12].

This study focuses on LQR and MPC design for a satellite attitude control system. The LQR is a well-known control technique for Linear Time-Invariant (LTI) systems, while MPC is a more recent approach that can handle nonlinearities and constraints. We design the LQR controller as an unconstrained optimization problem, we also discuss the effect of input saturation. We also design constrained-MPC problems and show that it is able to track the given references accurately. We also discuss the disturbance rejection performances of the controllers.

Outline of the paper is as follows: In the second section, we represent the equations of motion of a satellite that is required to model the attitude control system. We represent the nonlinear dynamics and kinematic equations and linearize them to obtain an LTI state-space system. The third section elaborates on the control system design to stabilize our minisatellite. The theory of LQR and MPC controllers is explained in detail. The fourth section consists of a simulation study that is used to simulate the controllers by using real satellite data. It also examines the effect of adding constraints for the LQR controller and constrained-MPC simulation results. We then compare the controllers in terms of disturbance rejection performances. The last section elaborates on the findings of the study and ideas that can be executed for ongoing studies.

II. EQUATIONS OF MOTION

A. Satellite Dynamics Model

For a satellite actuated by reaction wheels, dynamics equations can be written as:

$$\dot{\omega} = I_s^{-1} (-\omega \times I_s \omega - \omega \times I_{rw} \omega_{rw} - \tau_{rw} + \tau_e) \quad (1)$$

where $\omega \in \mathbb{R}^{3 \times 1}$ and $\omega_{rw} \in \mathbb{R}^{3 \times 1}$ are the satellite and reaction wheel angular velocity vectors respectively with respect to the inertial body frame, $\tau_{rw} \in \mathbb{R}^{3 \times 1}$ is the control torque vector provided by reaction wheels, $\tau_e \in \mathbb{R}^{3 \times 1}$ is the external torque vector acting on satellite, I_s and I_{rw} are the satellite and reaction wheel moment of inertia matrix, respectively. The skew-symmetric matrix for angular velocity to represent cross-product can be written as:

$$\omega_{\times}(\omega) \equiv \omega \times = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \quad (2)$$

By using (1), one can yield the angular velocity vector of a satellite by using torques acting on it as the input. It is also known that the reaction wheel torque can be calculated from the time derivative and opposite direction of the reaction wheel angular momentum:

$$\tau_{rw} = -\frac{d}{dt} (I_{rw} \omega_{rw}) \quad (3)$$

B. Satellite Kinematics Model

The Kinematics model can be represented by using quaternions [13], [14]:

$$\dot{q} = \frac{1}{2} \Omega(\omega) q \quad (4)$$

where:

$$\Omega = \Omega(\omega) = \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix} \quad (5)$$

Elements of the quaternion are:

$$q = \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} \quad (6)$$

where q_0 is the scalar and $\vec{g} \equiv (q_1, q_2, q_3)$ represents the vectorial elements of the unit quaternion. Equation (6) can be separated into two parts to separate vectorial and scalar elements of the quaternion [15]:

$$\begin{bmatrix} \dot{q} \\ \dot{q}_0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -\omega_{\times} g + q_4 I_{3 \times 3} \omega \\ -\omega^T g \end{bmatrix} \quad (7)$$

C. Combined Nonlinear Model

Let us define a state vector $\vec{x} = [\omega_1 \ \omega_2 \ \omega_3 \ q_0 \ q_1 \ q_2 \ q_3 \ \omega_{rw \ 1} \ \omega_{rw \ 2} \ \omega_{rw \ 3}]^T$. Equations (1) and (6) can be combined and represented as a nonlinear state space equation:

$$\dot{x} = \begin{bmatrix} \dot{\omega} \\ \dot{q} \\ \dot{q}_0 \\ \dot{\omega}_{rw} \end{bmatrix} = \begin{bmatrix} I_s^{-1} (-\omega \times I_s \omega - \omega \times I_{rw} \omega_{rw}) - \tau_{rw} + \tau_e \\ -\frac{1}{2} \omega_{\times} g + q_4 I_{3 \times 3} \omega \\ -\frac{1}{2} \omega^T g \\ -\tau_{rw} / I_{rw} \end{bmatrix} \quad (8)$$

D. Dimension Reduction of the Nonlinear Model

Yang showed that the quaternion kinematics model can be represented by using only three quaternions, namely the vector g . The reduced quaternion kinematics model can be represented as [16]:

$$\dot{g} = \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} Q(q_1, q_2, q_3) \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \quad (9)$$

where

$$Q \equiv \begin{bmatrix} q_0 & -q_3 & q_2 \\ q_3 & q_0 & -q_1 \\ -q_2 & q_1 & q_0 \end{bmatrix} \quad (10)$$

Using a reduced quaternion model, one can design a linear controller by using classical techniques since this model is fully controllable [17]. Dimension reduction is also another advantage whereas the disadvantage is the singularity of the system when the angle is $\pm\pi$ radians. Since the predicted operational angle is much smaller than 180 deg, the singularity will be no issue. To complete all the quaternions, the scalar element can be calculated by the relation:

$$q_0 = \sqrt{1 - q_1^2 - q_2^2 - q_3^2} \quad (11)$$

Additionally, the reaction wheel angular momentum element presented in dynamics equations can be neglected since it does not have a dramatic effect on the system dynamics. By applying this condition, (1) becomes:

$$\dot{\omega} = I_s^{-1} (-\omega \times I_s \omega - \tau_{rw} + \tau_e) \quad (12)$$

This condition also vanishes the reaction wheel angular velocity state (ω_{rw}). Finally, The simplified nonlinear state space model becomes:

$$\dot{x} = \begin{bmatrix} \dot{\omega} \\ \dot{g} \end{bmatrix} = \begin{bmatrix} I_s^{-1} (-\omega_{\times} I_s \omega - \tau_{rw} + \tau_e) \\ \frac{1}{2} Q \omega \end{bmatrix} \quad (13)$$

E. Linearized Satellite Equations

The nonlinear system can be linearized around an equilibrium point. Let us take the control torque, τ_{rw} as the control input u . Neglecting disturbances, nonlinear equations can be written as:

$$F(x, u) = \begin{bmatrix} I_s^{-1}(-\omega \times I_s \omega - u) \\ \frac{1}{2} Q \omega \end{bmatrix} \quad (14)$$

If one takes the equilibrium point as $x_0 = [0 \ 0 \ 0 \ 0 \ 0 \ 0]^T$, the linear state space equations become (please notice that $q_0 = 1$ with this conditions):

$$\begin{bmatrix} \dot{\omega} \\ \dot{g} \end{bmatrix} = \begin{bmatrix} 0_{3 \times 3} & 0_{3 \times 3} \\ \frac{1}{2} I_{3 \times 3} & 0_{3 \times 3} \end{bmatrix} \begin{bmatrix} \omega \\ g \end{bmatrix} + \begin{bmatrix} I_s^{-1} \\ 0_{3 \times 3} \end{bmatrix} u \quad (15)$$

$$= Ax + Bu$$

Since this linear state space system is controllable, classical control methods can be applied directly. Here, the number of states can be represented by n and the number of control inputs can be represented by m .

III. CONTROL SYSTEM DESIGN

A. LQR Controller

Linear Quadratic Regulator (LQR) controller is an optimal control method on tracking reference attitude for a satellite attitude control system. It relies on the minimization of a cost function that represents the control effort and tracking error. The cost function J can be constructed as [6]:

$$J = \int_0^\infty (x^T Q x + u^T R u) dt \quad (16)$$

Where $Q_{n \times n}$ and $R_{m \times m}$ are positive definite matrices that determine the weights of control input and tracking error. The control law that minimizes that cost function is determined as:

$$u = -Kx \quad (17)$$

where

$$K = R^{-1} B^T P \quad (18)$$

$P_{n \times n}$ is found by solving the Riccati differential equation [4]:

$$0 = PA + A^T P - Q + PBR^{-1}B^T P \quad (19)$$

The LQR control requires the system to be controllable. Thus, linearized state space equations given in (14) can be used directly.

B. Model Predictive Controller

Consider a discrete-time state-space model:

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) \end{aligned} \quad (20)$$

Where $A_{n \times n}$ is the state matrix $B_{m \times m}$ is the control matrix and $C_{p \times n}$ is the output matrix where p is the number of outputs.

Let us iterate the state space system given in (20) one step:

$$\begin{aligned} x(k+2) &= Ax(k+1) + Bu(k+1) \\ &= A(Ax(k) + Bu(k)) + Bu(k+1) \\ &= A^2x(k) + ABu(k) + Bu(k+1) \end{aligned} \quad (21)$$

If one continues the iterations, the prediction model becomes:

$$\begin{bmatrix} x(k+1) \\ x(k+2) \\ x(k+3) \\ \vdots \\ x(k+N_p) \end{bmatrix} = \begin{bmatrix} A \\ A^2 \\ A^3 \\ \vdots \\ A^{N_p} \end{bmatrix} x(k) + \begin{bmatrix} B & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ A^2B & AB & B & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A^{N_p-1}B & A^{N_p-2}B & A^{N_p-3}B & \cdots & B \end{bmatrix} \begin{bmatrix} u(k) \\ u(k+1) \\ u(k+2) \\ \vdots \\ u(k+N_p-1) \end{bmatrix} \quad (22)$$

which can be written in a compact form:

$$x_{\rightarrow} = Fx(k) + Hu_{\rightarrow} \quad (23)$$

Where N_p is the prediction horizon. By using (22), one can write the prediction equations to predict the ongoing measurements and control movements. Note that the prediction horizon is the same as the control horizon in (22). One can write a smaller control horizon to reduce the control horizon. Since there is no restriction on the computational process in this study, the control horizon is preferred same as the prediction horizon. It is also possible to predict the outputs by multiplying (22) with the output matrix, C . The cost function to be minimized for model predictive controller (MPC) structure is [8]:

$$\min J \quad \sum_{k=0}^{N_p-1} x_k^T Q x_k + u_k^T R u_k \quad (24)$$

subject to:

$$\begin{aligned} x_{k+1} &= Ax_k + Bu_k \quad k = 1, 2, \dots, N_p - 1 \\ u_{min} &\leq u_k \leq u_{max} \end{aligned}$$

where the matrices Q and R are the control block diagram for an MPC controller is given in Fig. 1. As shown in Fig. 1 there is an online optimizer inside of the MPC block. This optimizer is updated at every time step according to the data coming from the prediction model and produces proper control input by solving the optimization problem.

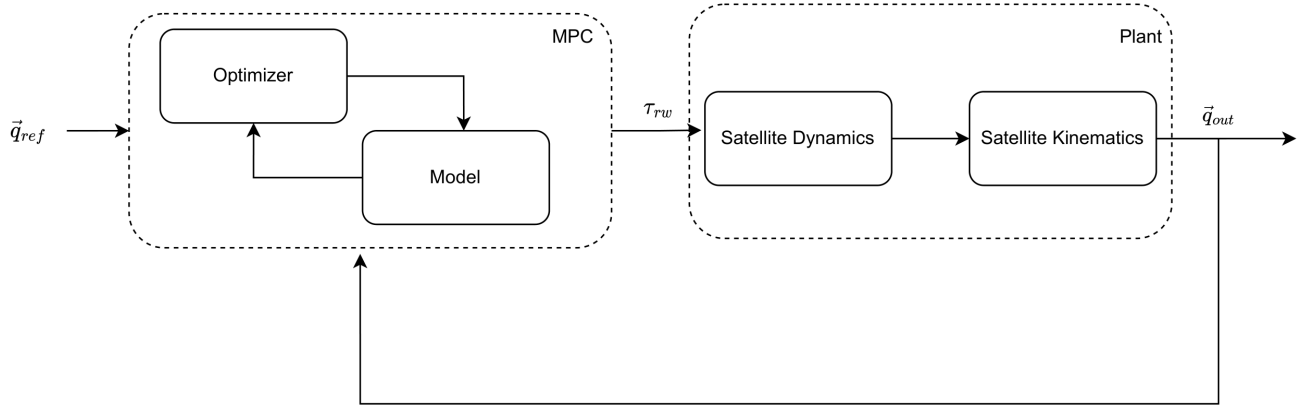


Fig. 1. MPC control diagram.

MPC has the capability of constraint handling. As shown by the cost function and constraint, it is possible to restrict the control movements with constraints that can represent the physical limits of reaction wheel motors. It is also possible to write constraints for states, and outputs as well. However, the constraint for only control movement is preferred for this study.

IV. SIMULATION STUDY

Reference tracking performance of the controllers can be evaluated by using simulation tools such as MATLAB/Simulink. Let us conduct a set of simulation scenarios to compare the controllers. Parameters of the simulation are given in Table I. UoSAT-12 satellite data is used for the construction moment of inertia matrix [18]. The reaction wheel control torque constraint value for the satellite is obtained from ref. [19].

TABLE I
SIMULATION PARAMETERS OF THE UoSAT-12 MINISATELLITE

Parameter	Symbol	Value	Unit
Satellite moment of inertia	I_s	$diag([40.45, 42.09, 40.36])$	kgm^2
Reaction wheel torque constraint	$\tau_{rw_{max}}$	± 0.02	Nm
Initial angular velocity	ω_0	$[0 \ 0 \ 0]$	rad/s
Initial quaternion elements	g_0	$[0 \ 0 \ 0]$	-

A. LQR Simulation

Firstly, the LQR controller was examined on the system. It is possible to find the optimal LQR gain given in (18) by inserting parameters from Table I, which gives the LQR gain as:

$$K_{LQR} = \begin{bmatrix} 6.4382 & 0 & 0 & 1 & 0 & 0 \\ 0 & 6.5643 & 0 & 0 & 1 & 0 \\ 0 & 0 & 6.4312 & 0 & 0 & 1 \end{bmatrix} \quad (25)$$

Eigenvalues of the closed loop system was calculated as:

$$eig(A - BK_{LQR}) = \begin{bmatrix} -0.080 + j0.078 \\ -0.080 - j0.078 \\ -0.080 + j0.078 \\ -0.080 - j0.078 \\ -0.078 + j0.076 \\ -0.078 - j0.076 \end{bmatrix} \quad (26)$$

which indicates that all the poles are on the left-hand side of the s-plane. Thus, the system is stable. A step reference was given to the system to test the output performance of the LQR controller. Reference value was set to 0.1, 0.05, 0.03 for g_1 , g_2 , and g_3 , respectively. Q and R matrices are set to $diag([2, 2, 2])$. Time response of the system are given in Fig. 2. As shown by Fig. 2, the satellite is now able to track the given reference accurately.

However, the classical LQR structure given in (16) do not have constraint handling capability. Thus, The response given in Fig. (2), does not include the saturation of reaction wheel torque, which is ± 0.02 Nm. To execute this, a saturation block was included to the system. The response of the system with and without saturation is given in Fig. 3. Additionally, the control input of the system with and without saturation is given in Fig. 4. Only g_1 and u_1 values are used to decrease the complexity of the graph.

As shown in Fig. 3, when the controller is saturated, settling time and overshoot increase as expected since the controller cannot give a torque higher than 0.02 Nm which is also shown in Fig. 4.

Saturated torque gives more realistic results. However, the classical LQR does not include constraints. With LQR, the optimization problem is solved as an unconstrained optimization problem. Thus, it can be said that the LQR solution with a saturation block is not an optimal solution. Moreover, including a saturation block means including a nonlinear parameter in the system. Thus, linearization of the system is required for further linear analysis. To include saturation into the optimization problem, more advanced techniques can be used such as model predictive controller.

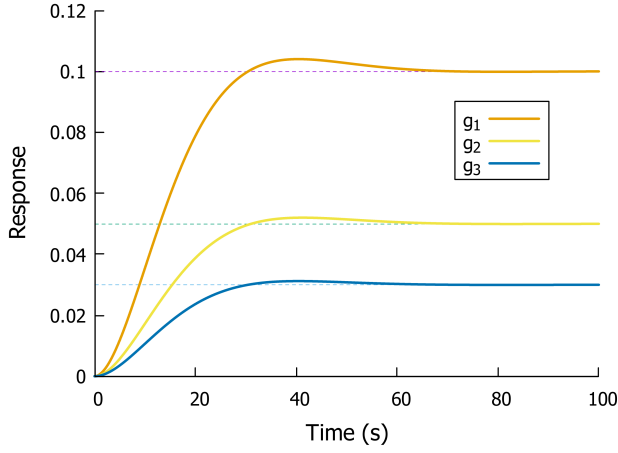


Fig. 2. Step response of the LQR controller.

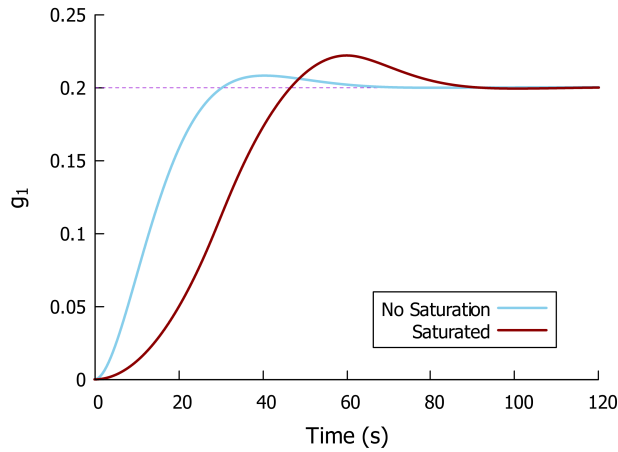


Fig. 3. Effect of controller saturation on g_1

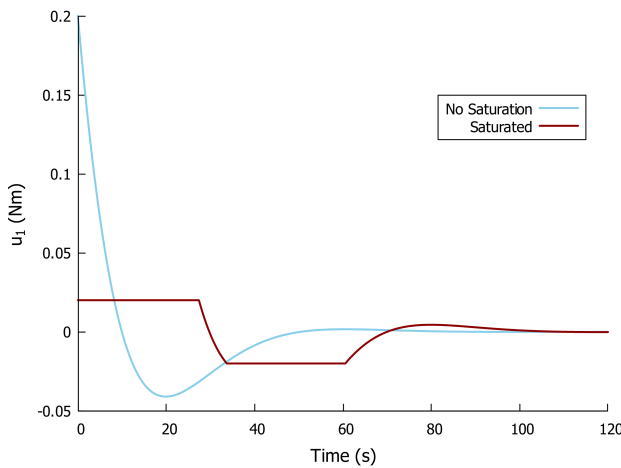


Fig. 4. Reaction wheel control torque with and without saturation.

B. MPC Simulation

Model Predictive Control may also include constrained and unconstrained solutions. The unconstrained MPC has an analytical solution [20].

Constrained-MPC transforms the optimization problem to a quadratic programming problem which can be solved using different techniques [21].

The MPC problem was written as in (24) and the input constraint is determined as $-0.02 \leq \tau_{rw} \leq 0.02$. Coefficients of the weight on control input, R , and the weight on reference tracking penalty, Q are determined as 0.1 and 50, respectively. Prediction and control horizons are chosen as 30. The sample time is 0.5 seconds. Reference quaternion elements are determined as $g_{ref} = [0.08, 0.05, 0.03]$. Under given MPC conditions, the step response of the quaternion elements is given in Fig 5.

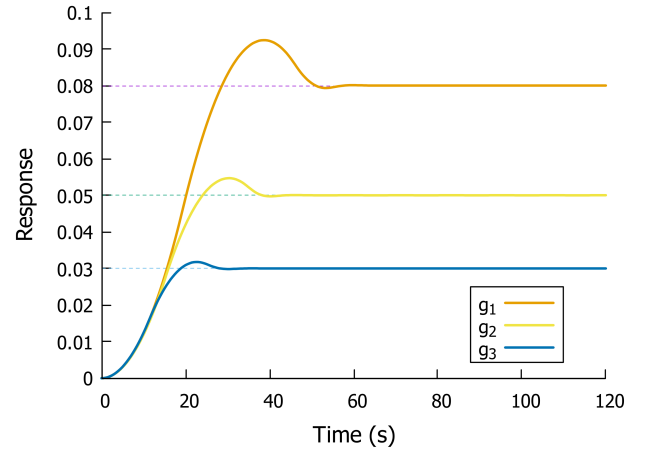


Fig. 5. Step response of the MPC controller.

As shown in Fig. 5, the controller is able to track given quaternion elements accurately. Settling time is smaller than LQR results given in Fig. 2. Additionally, the control input for the MPC controller is given in Fig. 6. As shown in Fig. 6, the controller is saturated properly within the constraints of the optimization problem.

C. Disturbance Rejection

To evaluate the disturbance rejection performance of the controllers, a step disturbance was given to the system. For reference, only the quaternion element, g_1 was taken 0.1 to reduce the complexity of the graph. The magnitude of the step disturbance was chosen as 0.01 and it is applied at the 70th second. Responses of the MPC and LQR controllers in the presence of offset disturbance in the output are given in Fig. 7.

V. CONCLUSION AND FUTURE WORK

The comparison of the two controllers, LQR and MPC, clearly demonstrates that the latter outperforms the former in terms of disturbance rejection performance, as illustrated by Fig. 7. The MPC controller shows the ability to effectively

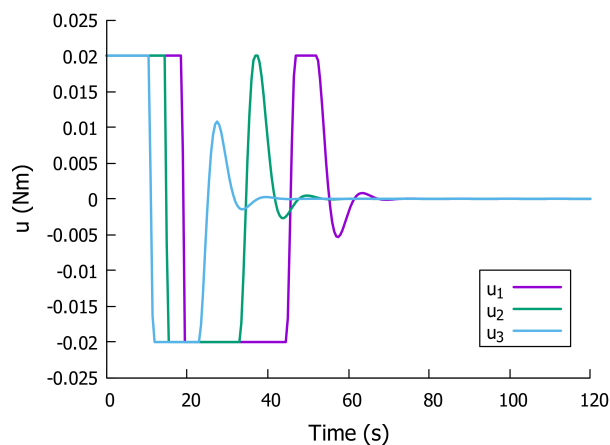


Fig. 6. Reaction wheel control torques for MPC controller.

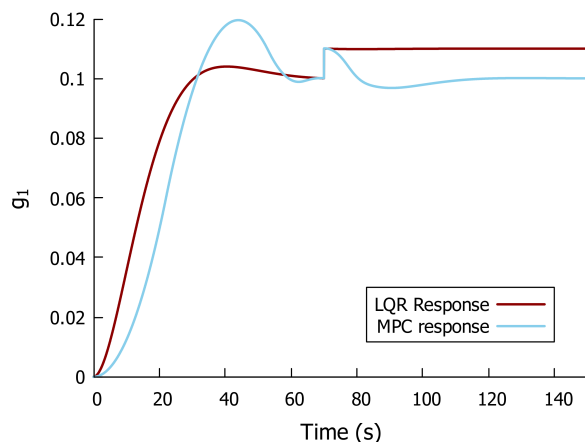


Fig. 7. Disturbance rejection performance of the controllers.

dampen constant disturbance, whereas the LQR controller fails to do so. Additionally, to gain a numerical insight, Root Mean Square Error (RMSE) values between controllers and reference were calculated after the disturbance was applied. RMSE for the LQR controller was found as 0.01 whereas RMSE for the MPC controller was calculated as 0.0029, which is exceedingly lower than LQR. Therefore, for the systems subject to disturbance, the MPC controller is the preferred choice, unless more advanced LQR methods that account for disturbance rejection are employed.

Future research on the comparison of LQR and MPC controllers could expand upon the use of observer models such as the Kalman filter. Additionally, incorporating disturbance models into the state space would enable the MPC controller to predict and account for disturbances, further improving its performance. Such investigations could provide valuable insights into the selection and optimization of control strategies for satellite attitude control and other similar applications.

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